



Predictive Model Plan

1. Model Logic (Generated with GenAI)

The predictive model is designed to forecast whether a customer is likely to become delinquent based on financial and behavioral attributes. Using GenAI, I conceptualized a Gradient Boosting model (e.g., XGBoost) due to its ability to capture complex patterns and interactions in customer data.

Model Workflow:

- **Data Ingestion:** Load customer-level data from the delinquency dataset.
- **Feature Selection:** Focus on key predictors such as Missed_Payments, Credit_Utilization, Debt_to_Income_Ratio, Credit_Score, and Account_Tenure.
- **Preprocessing:** Encode categorical variables, normalize numerical features, and handle missing values.
- **Model Training:** Use Gradient Boosting to learn patterns from training data.
- **Prediction Output:** Generate probability scores for delinquency and classify based on a threshold.
- **Evaluation:** Assess performance using metrics like AUC, F1 score, and fairness audits.

Pseudocode Example:

```
load_data("Delinquency_prediction_dataset.xlsx")
features = ["Missed_Payments", "Credit_Utilization", "Debt_to_Income_Ratio", "Credit_Score",
"Account_Tenure"]
```

```
X = preprocess(features)
y = target_variable("Delinquent_Account")
```

```
model = GradientBoostingClassifier()
model.fit(X_train, y_train)
predictions = model.predict_proba(X_test)
evaluate_model(predictions, y_test)
```

2. Justification for Model Choice

I selected **Gradient Boosting (XGBoost)** as the primary model due to its high predictive accuracy and ability to model non-linear relationships and feature interactions—essential for financial risk prediction. It handles outliers and missing data well, which are common in real-world financial datasets. While **Logistic Regression** offers transparency and ease of deployment, it assumes linearity and may miss complex behavioural patterns.

Gradient Boosting provides feature importance scores that support explainability, making it suitable for regulated environments like Geldium's. It aligns with business goals by offering precision, scalability, and compliance-readiness, while still allowing for stakeholder interpretation and auditability.

3. Evaluation Strategy

To ensure the model is both accurate and fair, I propose a multi-layered evaluation strategy:

Performance Metrics:

- **Accuracy:** Overall correctness of predictions.
- **Precision:** Measures how many predicted delinquents were correct.
- **Recall:** Measures how many actual delinquents were identified.
- **F1 Score:** Balances precision and recall.
- **AUC-ROC:** Evaluates model's ability to distinguish between classes.

Fairness & Bias Checks:

- **Demographic Parity:** Ensure equal positive prediction rates across groups (e.g., gender, income).
- **Equal Opportunity:** Check that true positive rates are consistent across demographics.
- **Disparate Impact Ratio:** Compare favourable outcomes across protected groups.
- **Calibration by Group:** Ensure predicted probabilities match actual outcomes across segments.

Bias Mitigation Techniques:

- Use preprocessing methods like SMOTE or reweighting to balance data.
- Apply fairness-aware algorithms or post-processing threshold adjustments.
- Monitor feature importance to detect over-reliance on sensitive attributes.

Ethical Considerations:

- Avoid reinforcing systemic biases.
- Ensure transparency in model decisions.
- Include human-in-the-loop review for borderline cases.
- Document model assumptions and limitations for audit purposes.